

TEAM MEMBER MEETINGS MINUTES

The Alchemist: AI-Powered Mineral Data Explorer

Project Name:	The Alchemist - AI-Powered Mineral Data Explorer
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Team Members:	Pratik Rauniyar, Sikha Chaudhary, Safal Shrestha, Bidhi Poudel

FALL SEMESTER 2025 (Sept 10 - Dec 1)

Phase 1: Project Kickoff, TA Setup, Tech Stack, Roles (Sept 10 - Oct 10)

Phase 1 - Meeting 1: Understanding Initial Setup & Project Overview

Date: Sept 10 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- Project overview and objectives
- Understanding the mineral data domain
- TA's initial setup and guidance
- Project timeline and deliverables
- Team introduction and expectations

Issues/Blockers:

None - Initial meeting

Action Items:

- Familiarize with project documentation
- Set up GitHub account and repository access
- Read about Mindat API
- Prepare questions for next meeting

Next Steps:

Next meeting: Technology stack selection and research

Phase 1 - Meeting 2: Technology Stack Selection & Research

Date: Sept 21 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- Review FastAPI vs other backend frameworks
- React + Vite frontend discussion
- LangChain and LangGraph evaluation
- Azure OpenAI vs other LLM providers
- Supabase for authentication and database
- Deployment platforms: Render and Vercel
- MCP (Model Context Protocol) research

Issues/Blockers:

Need to finalize on exact versions and compatibility

Action Items:

- Create tech stack documentation
- Install and test FastAPI locally
- Set up Node.js and React environment
- Research LangGraph tutorials
- Create architecture diagram

Next Steps:

Next meeting: Roles assignment and git workflow setup

Phase 1 - Meeting 3: Roles Assignment, Git Workflow & Contribution Rules

Date: Oct 1 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- Assigning developer roles (Frontend, Backend, DevOps)
- Git branching strategy (main, test, individual branches)
- Pull request and code review process
- Contribution guidelines and commit messages
- Local development environment setup plan
- Repository structure and folder organization
- Communication protocols

Issues/Blockers:

Ensuring everyone has proper GitHub permissions

Action Items:

- Create git workflow documentation
- Set up individual branches for each team member
- Configure GitHub protection rules

- Schedule local environment setup sessions
- Create coding standards document

Next Steps:

Next phase: Infrastructure and API configuration

Phase 2: Infrastructure & Configuration Setup (Oct 11 - Nov 10)

Phase 2 - Meeting 1: Mindat API Configuration & Environment Setup

Date: Oct 11 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- Mindat API documentation review
- API key generation and management
- API endpoint exploration (40+ endpoints)
- Query parameters and filtering options
- Authentication and rate limiting
- Testing Mindat API locally
- Environment variables setup (.env files)

Issues/Blockers:

API key security and access control

Action Items:

- Obtain Mindat API keys
- Create .env template file
- Test basic API calls using cURL/Postman
- Document API endpoints usage
- Set up environment variable management

Next Steps:

Next meeting: Supabase setup and database configuration

Phase 2 - Meeting 2: Supabase Setup & Database Configuration

Date: Oct 23 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- Supabase project creation and setup
- PostgreSQL database configuration
- Database schema design and tables
- User and Profile tables
- Session and Message tables
- Authentication integration (OAuth, email/password)
- Connection string and security keys

Issues/Blockers:

Database relationship design complexity

Action Items:

- Create Supabase project
- Set up PostgreSQL tables
- Configure Row Level Security (RLS)
- Generate connection strings
- Test database connections locally
- Document database schema

Next Steps:

Next meeting: API endpoints definition and local dev setup

Phase 2 - Meeting 3: API Endpoints Definition & Local Development Setup

Date: Nov 3 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- Defining FastAPI routes (/api/auth, /api/sessions, /api/agent)
- Request/response models and validation
- CORS configuration for frontend
- Error handling and status codes
- Local development environment on each computer
- Virtual environment setup and dependency installation
- Database migration strategy

Issues/Blockers:

Ensuring all team members have identical local environments

Action Items:

- Each member: Set up virtual environment
- Install Python dependencies (pip install -r requirements.txt)
- Install Node.js dependencies (npm install)
- Create local .env files

- Test database connection
- Run FastAPI server locally
- Verify Supabase connectivity

Next Steps:

Next phase: Backend development begins

Phase 3: Backend Routes Development (Nov 11 - Nov 14)

Phase 3 - Meeting 1: Backend Routes & Database Models Implementation

Date: Nov 11 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- Implementing auth endpoints (register, login, logout)
- User CRUD operations
- Session management endpoints
- Message storage and retrieval
- SQLAlchemy ORM models
- Database migration setup
- Testing auth flow
- Supabase auth integration with FastAPI

Issues/Blockers:

Coordinating database migrations across team

Action Items:

- Implement basic auth routes
- Create SQLAlchemy models
- Test auth endpoints with Postman
- Verify database updates
- Document API endpoints
- Handle edge cases and errors

Next Steps:

BREAK: Nov 15-25 (No meetings)

***** THANKSGIVING BREAK: Nov 15 - Nov 25 (No meetings) *****

Phase 4: LangChain & LangGraph Agent System (Nov 26 - Dec 1)

Phase 4 - Meeting 1: LangGraph Architecture & Agent Planning

Date: Nov 26 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- LangGraph state management and workflow
- Supervisor agent design and routing logic
- Planning specialized agents (General, Geomaterial, Locality, Plot)
- Agent state transitions
- Tool integration architecture
- Testing agent workflows
- Integration with FastAPI

Issues/Blockers:

Complexity of multi-agent coordination

Action Items:

- Design detailed agent architecture diagram
- Create agent prompts and instructions
- Set up LangGraph project structure
- Begin Supervisor agent implementation
- Plan agent testing strategy
- Document agent interactions

Next Steps:

WINTER BREAK: Dec 10 - Jan 18 (No meetings)

*** *WINTER BREAK: Dec 10 - Jan 18 (No meetings)* ***

SPRING SEMESTER 2026 (Feb 5 - Apr 29)

Phase 5: Frontend Development & UI/UX Design (Feb 5 - Mar 5)

Phase 5 - Meeting 1: UI/UX Design & Component Architecture

Date: Feb 5 Time: 5:00pm

Attendees:

All team members present, focus on frontend developers

Topics Discussed:

- UI/UX design mockups and prototypes
- Dark theme design specifications
- Chat interface layout
- Component hierarchy and structure
- Responsive design for mobile/tablet/desktop
- Icon selection and styling
- Color palette and typography

Issues/Blockers:

Ensuring design consistency across components

Action Items:

- Create detailed UI mockups
- Design component library structure
- Set up Figma/design tool
- Define CSS/Bootstrap structure
- Create reusable component patterns
- Review and approve designs

Next Steps:

Next meeting: React component development

Phase 5 - Meeting 2: React Components Development - Part 1

Date: Feb 16 Time: 5:00pm

Attendees:

All team members present, focus on frontend developers

Topics Discussed:

- Chat interface main component
- MessageBubble component for message display
- ChatInput component for user input
- Sidebar component for chat history
- Authentication pages (SignIn, Signup)
- Component state management
- Props and data flow

Issues/Blockers:

Managing component dependencies and prop drilling

Action Items:

- Create Chat.jsx main page
- Implement MessageBubble component

- Build ChatInput component
- Create Sidebar with chat list
- Implement SignIn/Signup pages
- Test component rendering
- Handle component state updates

Next Steps:

Next meeting: Advanced components and API integration

Phase 5 - Meeting 3: React Components Development - Part 2 & API Integration

Date: Feb 27 Time: 5:00pm

Attendees:

All team members present, focus on frontend developers

Topics Discussed:

- VegaChart component for visualization
- DataTable component for results
- Settings and Profile modals
- React Router setup and navigation
- Axios API client configuration
- Frontend-backend API calls
- Error handling and loading states

Issues/Blockers:

Ensuring proper API response handling

Action Items:

- Implement VegaChart visualization
- Create DataTable component
- Build Settings and Profile modals
- Configure Axios interceptors
- Implement API call handling
- Add loading and error states
- Test frontend with backend API

Next Steps:

Next phase: MCP development and tool integration

Phase 6: MCP Server & Tool Development (Mar 6 - Mar 14)

Phase 6 - Meeting 1: MCP Server Setup & Tool Definitions

Date: Mar 6 Time: 5:00pm

Attendees:

All team members present, focus on backend developers

Topics Discussed:

- FastMCP server setup and configuration
- MCP tool registration framework
- Tool definitions: collect_geomaterials
- Tool definitions: collect_localities
- Tool definitions: data profiling tools
- Tool input/output schemas
- Tool documentation

Issues/Blockers:

Ensuring proper tool definition structure

Action Items:

- Set up FastMCP server
- Create tool registration files
- Implement collect_geomaterials tool
- Implement collect_localities tool
- Create data profiling tool
- Test tools with sample data
- Document tool parameters

Next Steps:

Next meeting: MCP integration with backend

Phase 6 - Meeting 2: MCP Integration with Backend & Agent Testing

Date: Mar 11 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- Connecting MCP server to FastAPI backend
- MCP client configuration
- Tool invocation from agents
- Tool response handling
- Error handling and fallbacks
- Testing tool execution
- Performance testing

Issues/Blockers:

Debugging MCP communication issues

Action Items:

- Configure MCP_SERVER_URL in backend
- Test tool invocation
- Verify tool responses
- Handle error cases
- Optimize tool performance
- Document integration
- Prepare for agent integration

Next Steps:

SPRING BREAK: Mar 15-25 (No meetings)

*** *SPRING BREAK: Mar 15 - Mar 25 (No meetings)* ***

Phase 7: Agent Development & Full Integration (Mar 26 - Apr 20)

Phase 7 - Meeting 1: Specialized Agents Development - Part 1

Date: Mar 26 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- General Agent implementation (greetings, FAQs)
- Geomaterial Collector Agent development
- Agent prompts and system messages
- Tool invocation logic
- Response parsing and formatting
- Agent state management
- Testing individual agents

Issues/Blockers:

Coordinating agent development across team

Action Items:

- Implement General Agent
- Develop Geomaterial Collector Agent
- Create agent prompts
- Test agent responses
- Debug routing issues
- Optimize agent performance
- Document agent behavior

Next Steps:

Next meeting: More agents and visualization

Phase 7 - Meeting 2: Specialized Agents Development - Part 2

Date: Apr 6 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- Locality Collector Agent implementation
- Vega Plot Generator Agent
- Visualization specification generation
- Chart type selection logic
- Data aggregation for visualizations
- Agent testing and debugging
- Agent integration with Supervisor

Issues/Blockers:

Managing complex visualization generation

Action Items:

- Develop Locality Collector Agent
- Implement Plot Generator Agent
- Create Vega-Lite specs
- Test visualization generation
- Handle edge cases
- Optimize visualization queries
- Integrate with Supervisor

Next Steps:

Next meeting: Full system testing and CI/CD

Phase 7 - Meeting 3: Full System Integration, Testing & CI/CD Pipelines

Date: Apr 14 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- End-to-end workflow testing
- Testing all query types
- User journey testing

- Performance benchmarking
- CI/CD pipeline setup (GitHub Actions)
- Automated testing implementation
- Deployment pipeline configuration
- Load testing and optimization

Issues/Blockers:

Ensuring system stability and performance

Action Items:

- Run comprehensive integration tests
- Test complete user flows
- Performance profiling
- Set up GitHub Actions CI/CD
- Configure automated testing
- Optimize slow queries
- Prepare deployment scripts
- Document testing procedures

Next Steps:

Next phase: Final preparation for design expo

Phase 8: Final Design Expo Preparation (Apr 21 - Apr 29)

Phase 8 - Meeting 1: Design Expo Preparation & Demo Walkthrough

Date: Apr 21 Time: 5:00pm

Attendees:

All team members present

Topics Discussed:

- Design expo presentation planning
- Demo script and walkthrough
- Showcase key features and capabilities
- Agent capability demonstration
- UI/UX presentation
- Technical architecture overview
- Final optimizations and polish
- Handling demo edge cases

Issues/Blockers:

None - Final phase

Action Items:

- Prepare presentation slides
- Create demo dataset
- Test demo workflow end-to-end
- Prepare backup demo scenarios
- Final code cleanup and documentation
- Deploy to production
- Set up demo environment
- Brief team on presentation

Next Steps:

DESIGN EXPO DAY: April 29 - Showcase project to judges and attendees